ITA0612 – Machine Learning

CO2 – ASSIGNMENT II – SDG QUESTIONS

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Question

Design the input and output representation of a Neural Network intended to predict student dropout risk in under-resourced schools. Justify your feature encoding choices. (SDG 4 – Quality Education)

1. Problem Understanding

Objective

The objective of this project is to design the input and output representation of an Artificial Neural Network (ANN) that predicts the dropout risk of students studying in under-resourced schools. The system classifies each student into one of three risk categories — Low, Medium, or High — using academic, attendance, socio-economic, and behavioural indicators, so that schools can identify and support at-risk students before they leave the education system.

Problem Statement

Student dropout is a persistent problem in under-resourced schools, where poor attendance, low family income, lack of digital access, and limited parental support combine to push vulnerable students out of education. Schools typically hold this information in scattered administrative records and rarely combine it into a single early-warning signal. A Neural Network can learn the non-linear relationships between these factors and the eventual dropout outcome, producing a risk score that teachers and counsellors can act on well before a student actually leaves school.

Importance of Predicting Student Dropout

- Enables early intervention through counselling, remedial classes, or financial support before a student disengages completely.
- Helps school administrators allocate limited resources (scholarships, meal programmes, tutoring) to the students who need them most.
- Reduces long-term social and economic costs associated with an incomplete education.
- Provides measurable indicators that policymakers can use to evaluate the effectiveness of retention programmes.

How AI and Neural Networks Help Schools Identify At-Risk Students

Traditional rule-based flags (for example, "attendance below 60%") only capture one dimension of risk and generate many false alarms. A Neural Network instead learns weighted, non-linear combinations of dozens of academic, behavioural, and socio-economic features simultaneously. It can recognise, for instance, that a student with moderate attendance but very low family income, no internet access, and negative teacher feedback is at materially higher risk than the numbers would suggest in isolation. Because the network outputs a probability for each risk category, schools receive a ranked, continuously-updated watch-list rather than a single rigid cut-off, which makes early intervention far more targeted and effective.

Relevance to SDG 4 – Quality Education

This project directly supports United Nations Sustainable Development Goal 4 (Quality Education), which calls for inclusive and equitable quality education and the promotion of lifelong learning opportunities for all. By predicting dropout risk early — especially in under-resourced schools where support staff and counsellors are scarce — the model helps ensure that vulnerable children are not left behind, supporting Target 4.1 (free, equitable, and quality primary and secondary education) and Target 4.5 (eliminating gender and socio-economic disparities in education).

Mitchell's Criteria

Component	Description
Task (T)	Predict the dropout risk of a student — classified as Low, Medium, or High.
Experience (E)	Historical student records covering attendance, academic performance, socio-economic status, family background, and behavioural indicators.

Component	Description
Performance (P)	Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC evaluated on held-out test data.

2. Data Collection and Preparation

Dataset

Dataset Name: student_dropout.csv

Number of Records: 1000 (synthetically generated to reflect realistic under-resourced-school conditions)

Attribute Table

Attribute	Description	Data Type
Student_ID	Unique student identifier	String
Age	Student age (11–17 years)	Numeric
Gender	Male / Female	Categorical
Grade_Level	Current grade (6–12)	Numeric
Attendance_Percentage	Percentage of school days attended	Numeric
Previous_Exam_Score	Score obtained in the previous examination (%)	Numeric
Current_GPA	Current grade point average (0–4 scale)	Numeric
Family_Income	Annual family income	Numeric
Parent_Education	Highest education level of parents	Ordinal
Distance_to_School	Distance from home to school (km)	Numeric
Internet_Access	Whether the household has internet access	Binary
Device_Availability	Whether a learning device is available at home	Binary
Scholarship_Status	Whether the student receives a scholarship	Binary
Midday_Meal_Support	Whether the student receives midday meal support	Binary
Extracurricular_Participation	Participation in extracurricular activities	Binary
Behavioral_Score	Behavioural conduct score (0–100)	Numeric
Health_Status	Good / Average / Poor	Ordinal
Learning_Disability	Whether a diagnosed learning disability is present	Binary
Teacher_Feedback	Positive / Neutral / Negative	Ordinal
Homework_Completion_Rate	Percentage of homework completed on time	Numeric
Dropout_Risk	Target label: Low / Medium / High	Categorical (Target)

Sample Records

A representative sample of the generated dataset is shown below:

Student_ID	Age	Gender	Attendance %	GPA	Family Income	Dropout Risk
STU0001	17	Male	58.2	1.72	24,150	High
STU0002	14	Female	89.4	3.21	31,900	Low

Student_ID	Age	Gender	Attendance %	GPA	Family Income	Dropout Risk
STU0003	15	Male	82.1	2.65	12,400	Low
STU0004	17	Female	55.7	1.98	9,300	High
STU0005	13	Male	63.0	2.10	15,700	High

Data Preprocessing

The raw dataset was cleaned and transformed through the following steps before being fed to the Neural Network:

Missing Value Handling

About 2% of values in Previous_Exam_Score, Family_Income, Behavioral_Score, and Teacher_Feedback were missing, simulating real-world data collection gaps. Numeric columns were imputed with the column median (robust to outliers such as very high family incomes), and the categorical column was imputed with the column mode.

One-Hot Encoding

Gender was one-hot encoded into two binary columns (Gender_Male, Gender_Female) because it is a nominal variable with no inherent order — assigning it a single ordinal number would falsely imply a ranking between categories.

Label / Ordinal Encoding

Parent_Education, Health_Status, and Teacher_Feedback were ordinal-encoded (e.g., No Formal Education = 0 … Graduate = 4) because these categories have a natural order, which lets the network use the encoded value's magnitude as meaningful information rather than treating each category as unrelated.

Normalization and Standardization

Continuous features with roughly bell-shaped distributions (Age, Grade_Level, Attendance_Percentage, Previous_Exam_Score, Current_GPA, Behavioral_Score, Homework_Completion_Rate) were standardized (zero mean, unit variance) using z-score scaling. Highly skewed features (Family_Income, Distance_to_School) were instead Min-Max scaled to the range [0, 1], which compresses extreme values without assuming a normal distribution.

Train-Test Split and Feature Scaling

The processed dataset was split into 80% training data (800 records) and 20% testing data (200 records) using stratified sampling so that the proportion of Low, Medium, and High risk students is preserved in both sets. All scalers (Min-Max, Standard) were fit only on the training data and then applied to the test data, to avoid information leakage from the test set into the training process.

Preprocessing Pipeline

Figure 2.1. Data preprocessing pipeline from raw CSV to model-ready tensors.

3. Design of Neural Network Input and Output Representation

Every categorical, ordinal, and numeric attribute in the dataset must be converted into a fixed-length numeric vector before it can be presented to the Neural Network. This section explains, feature by feature, why a particular encoding method was chosen, and how the individual encodings combine into the final input vector.

Feature Encoding Justification

Feature	Encoding Method	Justification
Age, Grade_Level, Attendance %, Exam Score, GPA, Behavioral Score, Homework %	Standardization (Z-score)	Continuous, roughly bell-shaped values; standardization keeps gradients stable during backpropagation.
Family_Income, Distance_to_School	Min-Max Scaling	Highly skewed / long-tailed values; Min-Max keeps them bounded in [0,1] without assuming normality.
Gender	One-Hot Encoding	Nominal category with no order — one-hot avoids implying a false ranking.
Parent_Education, Health_Status, Teacher_Feedback	Ordinal Encoding	Categories have a natural order, so a single increasing integer preserves that order for the network.
Internet_Access, Device_Availability, Scholarship_Status, Midday_Meal_Support, Extracurricular_Participation, Learning_Disability	Binary Encoding (0/1)	Simple Yes/No attributes; a single binary flag is the most compact, information-preserving representation.

Input Layer Design

After encoding, the feature vector for each student consists of: 7 standardized numeric features, 2 Min-Max scaled numeric features, 2 one-hot Gender columns, 3 ordinal-encoded columns, and 6 binary columns — giving a total of 21 input neurons.

Encoding Group	Number of Neurons
Standardized numeric features	7
Min-Max scaled numeric features	2
One-hot Gender	2
Ordinal-encoded features	3
Binary-encoded features	6
Total Input Neurons	21

Output Layer Design

The output layer has three neurons, one for each dropout-risk class: Low Risk, Medium Risk, and High Risk. The output layer uses the Softmax activation function, which converts the three raw output scores (logits) into a probability distribution that sums to 1. The class with the highest probability is taken as the network's prediction. Softmax is preferred over independent sigmoid units here because the three risk categories are mutually exclusive — a student belongs to exactly one class — and Softmax explicitly models that competition between classes.

Neural Network Architecture Diagram

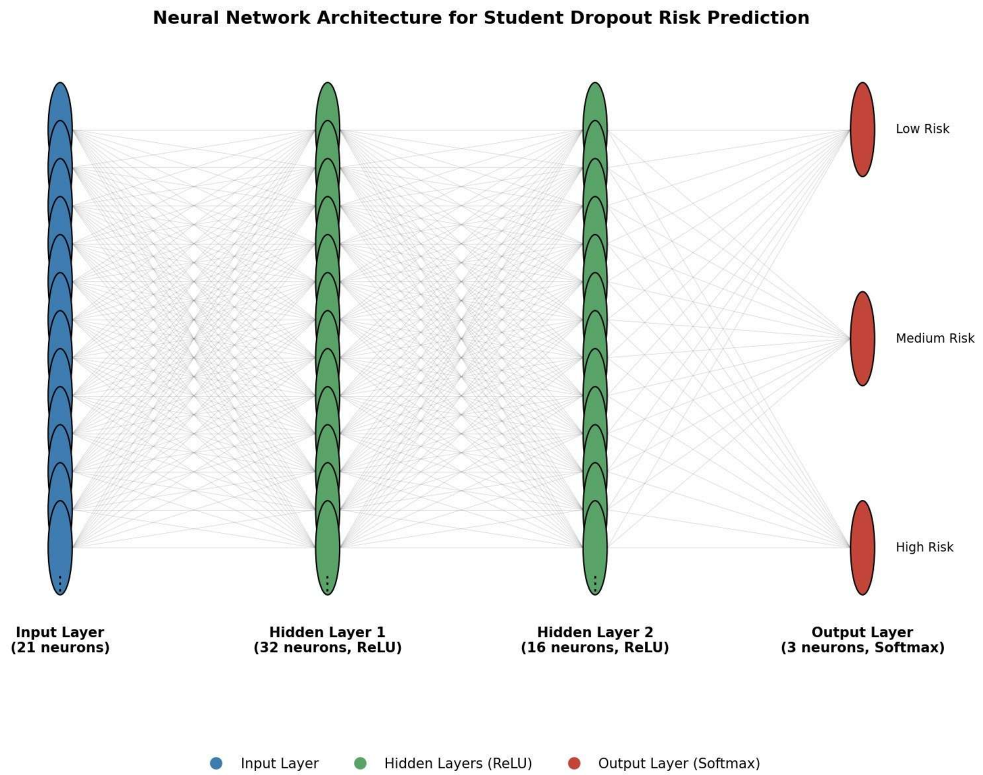

Figure 3.1. ANN architecture: 21 input neurons → 32 → 16 hidden neurons (ReLU) → 3 output neurons (Sokmax).

Input-Output Representation Diagram

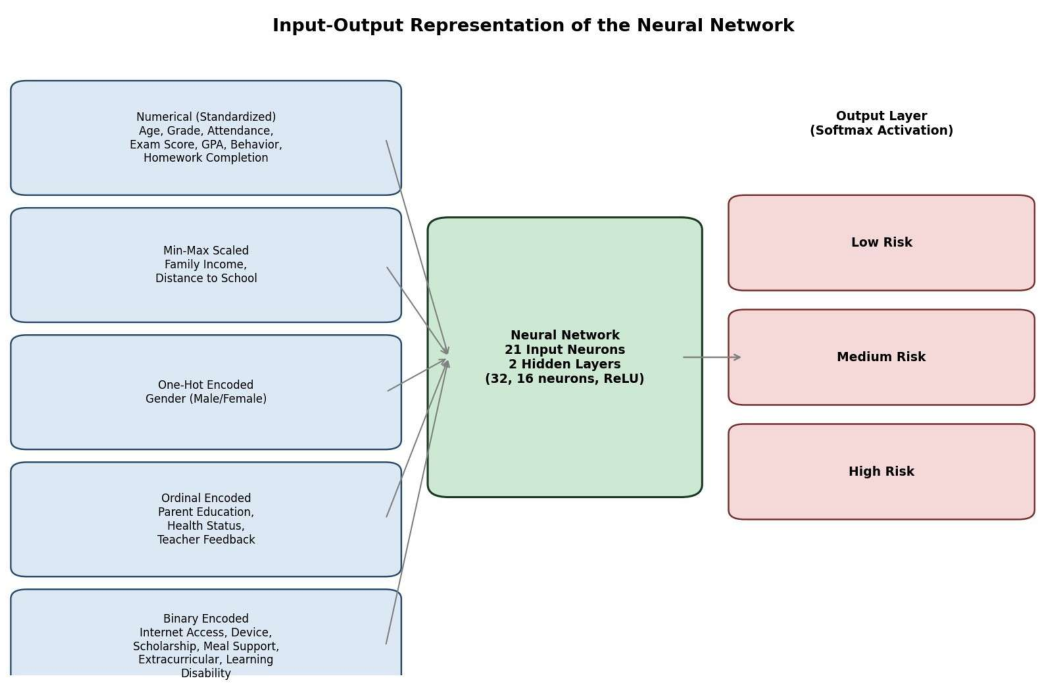

Figure 3.2. How each encoded feature group feeds into the network and maps to the three risk outputs.

4. NumPy and Pandas Analysis

Before model training, exploratory analysis was performed on the raw dataset using NumPy for numeric computation and Pandas for group-wise and tabular analysis.

Summary Statistics

Metric	Value
Average Attendance	78.05%
Average Current GPA	2.58 / 4.0
Average Family Income	19,683.74

These averages were computed with NumPy (numpy.mean) directly on the underlying arrays. An average attendance of 78% with a standard deviation of roughly 15 points indicates a wide spread — many students cluster near full attendance, but a meaningful tail sits well below 60%, which is exactly the group the model needs to flag early.

Dropout Risk Distribution

Risk Category	Number of Students	Percentage
Low	550	55.0%
Medium	270	27.0%
High	180	18.0%

The class distribution is imbalanced, which is realistic — most students are not at risk — but this also means the model must be evaluated with metrics beyond raw accuracy (see Section 7), since a naive model that always predicts "Low" would already be correct 55% of the time.

Average Attendance by Gender

Gender	Average Attendance (%)
Female	77.63
Male	78.44

This Pandas group-by analysis (df.groupby("Gender")["Attendance_Percentage"].mean()) shows only a small gap (under one percentage point) between genders in this synthetic dataset, suggesting attendance-driven risk is not strongly gender-dependent here — though this should always be verified against real school data, where such gaps can be more pronounced.

Correlation Matrix

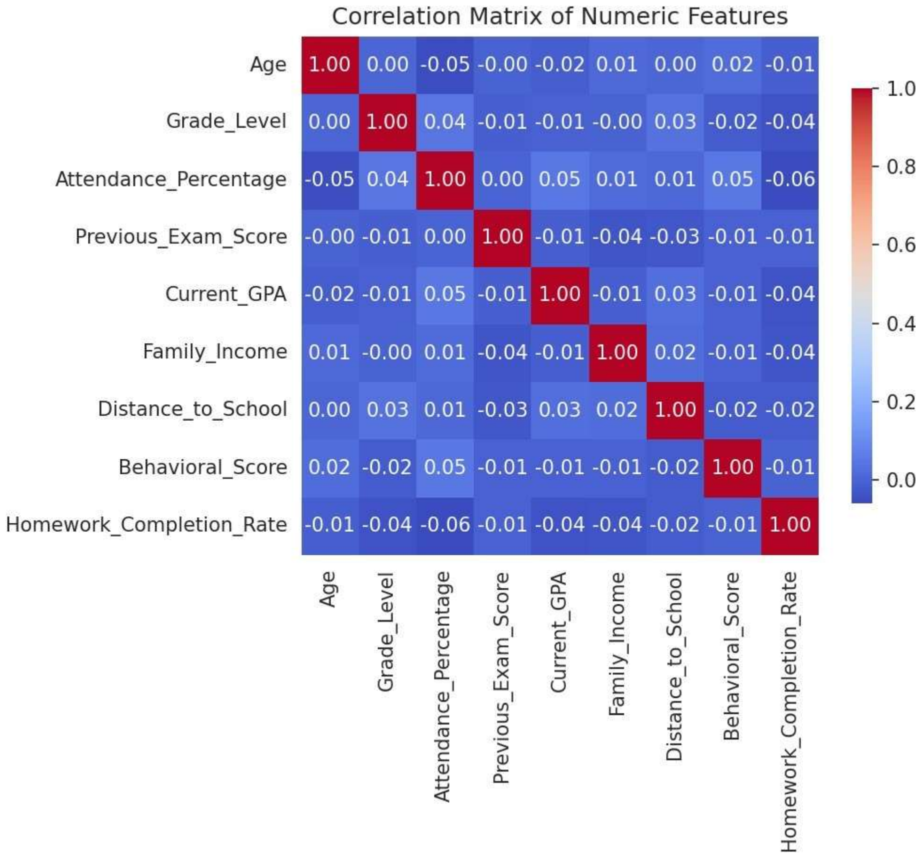

Figure 4.1. Pearson correlation matrix of the numeric features.

The heatmap shows the expected relationships: Attendance_Percentage correlates positively with Current_GPA and Previous_Exam_Score, while Family_Income and Distance_to_School show weaker, mostly independent relationships with academic performance — confirming that dropout risk is driven by a combination of factors rather than any single dominant variable.

Top Important Features

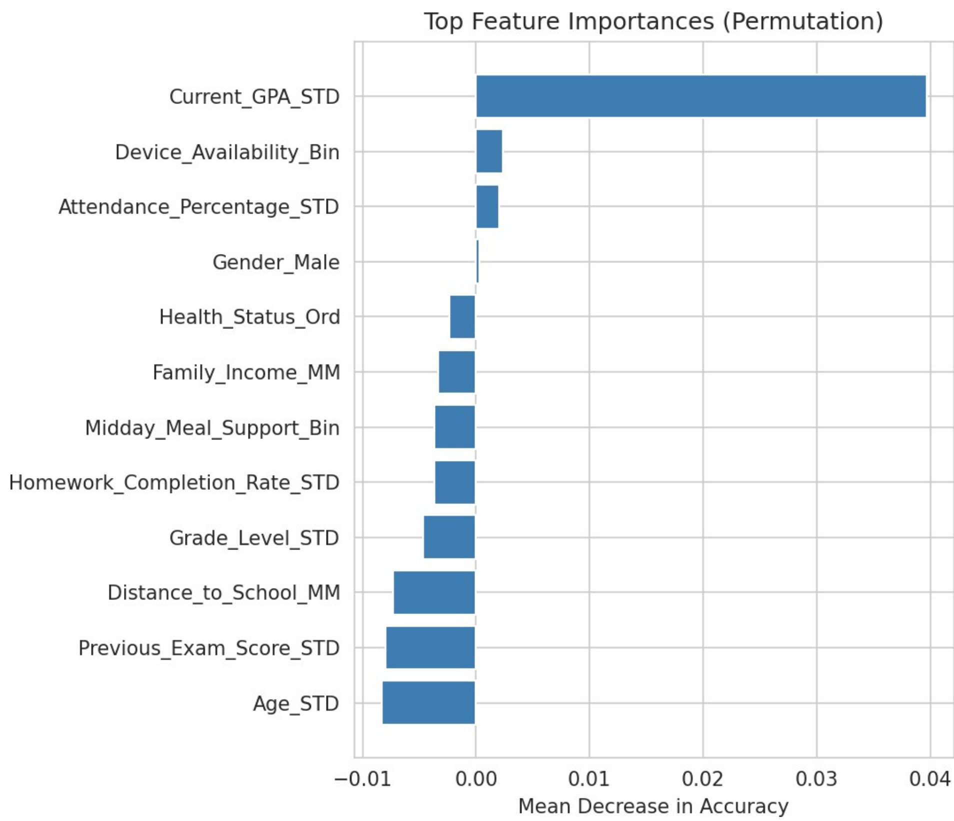

Figure 4.2. Permutation feature importance from the trained ANN (mean decrease in accuracy when a feature is shuffled).

Permutation importance measures how much test accuracy drops when a single feature's values are randomly shuffled, breaking its relationship with the target. Current_GPA and Attendance_Percentage emerge as the most influential features, consistent with educational research showing that in-school academic performance is the strongest short-term predictor of dropout, ahead of longer-acting socio-economic factors such as family income.

5. Visualization and Interpretation

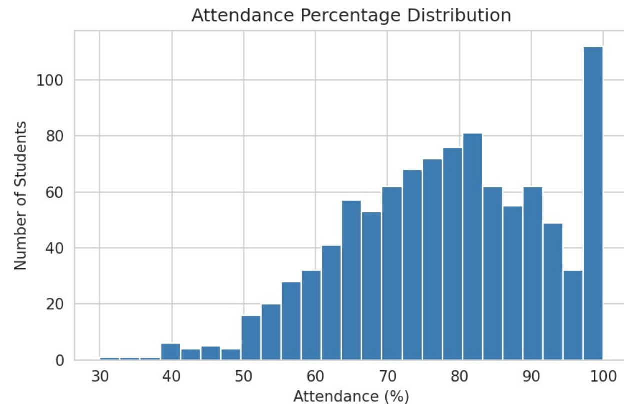

Figure 5.1. Attendance Percentage Distribution.

Most students attend between 65% and 95% of school days, with a visible left tail below 50% — this tail is the group most likely to fall into the High risk category.

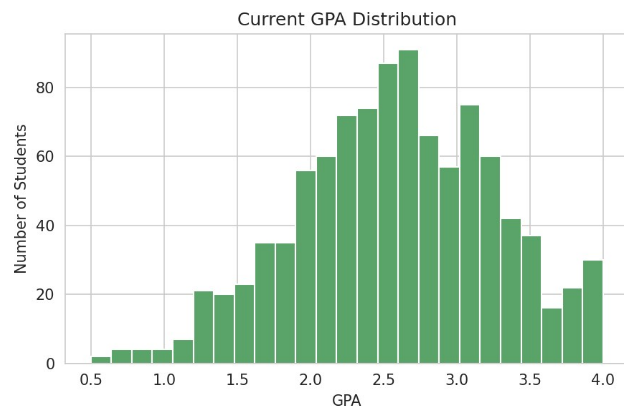

Figure 5.2. Current GPA Distribution.

GPA is roughly bell-shaped around 2.6, with a spread wide enough that a meaningful number of students sit below 1.5, a common academic-risk threshold.

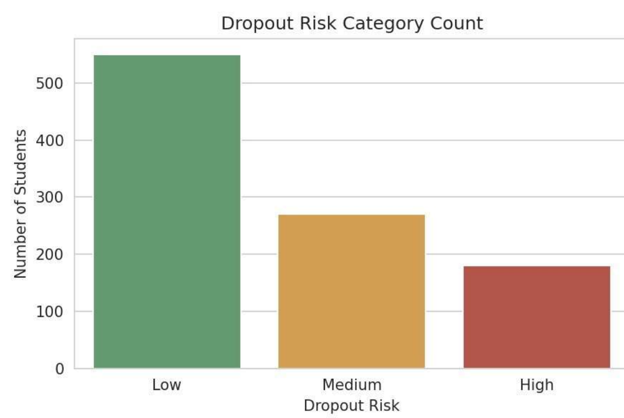

Figure 5.3. Dropout Risk Category Count.

Confirms the class imbalance noted in Section 4 — Low risk students outnumber High risk students roughly 3 to 1.

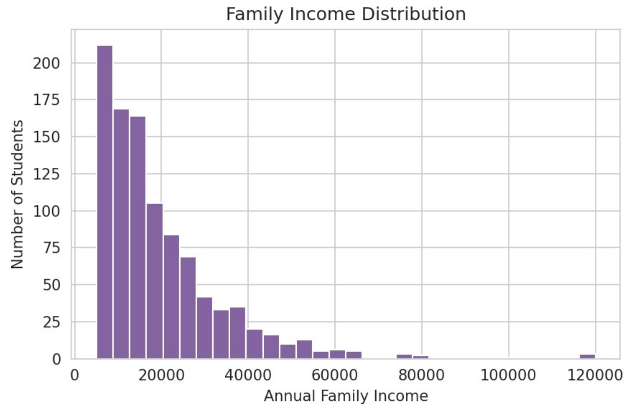

Figure 5.4. Family Income Distribution.

Income is right-skewed, typical of under-resourced communities: a large cluster of low-income families with a long tail of higher earners, which justifies the Min-Max scaling choice made in Section 2.

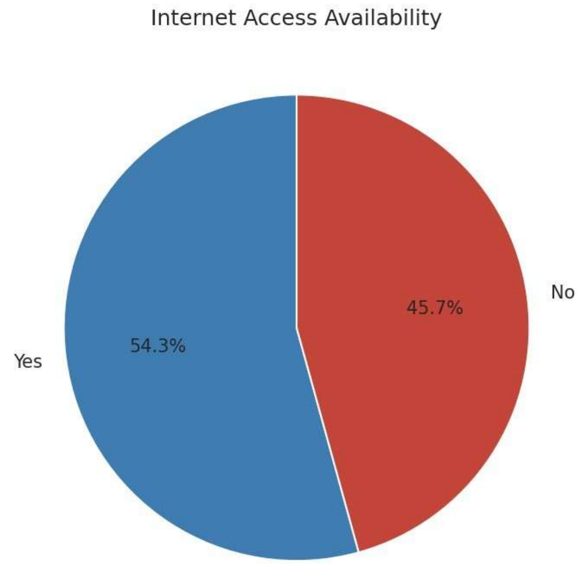

Figure 5.5. Internet Access Availability.

Roughly 45% of students lack home internet access — a substantial digital-divide segment directly relevant to remote or blended learning capability.

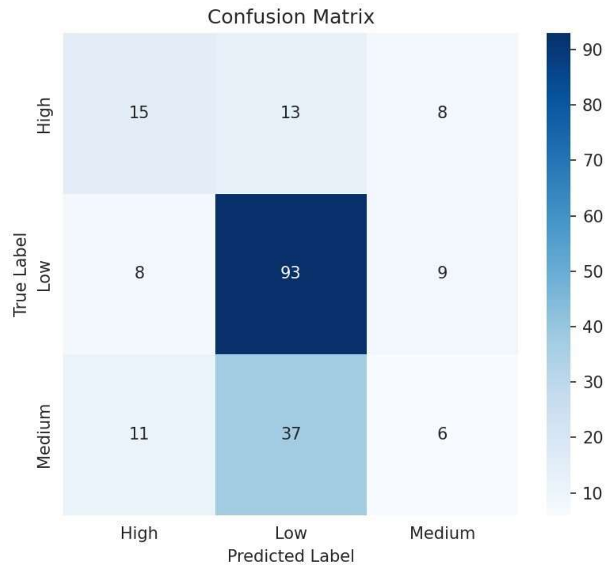

Figure 5.6. Confusion Matrix on the test set.

The diagonal dominates for the Low class; most confusion happens between the Medium and Low/High classes, which is expected since Medium risk sits on the boundary between the other two.

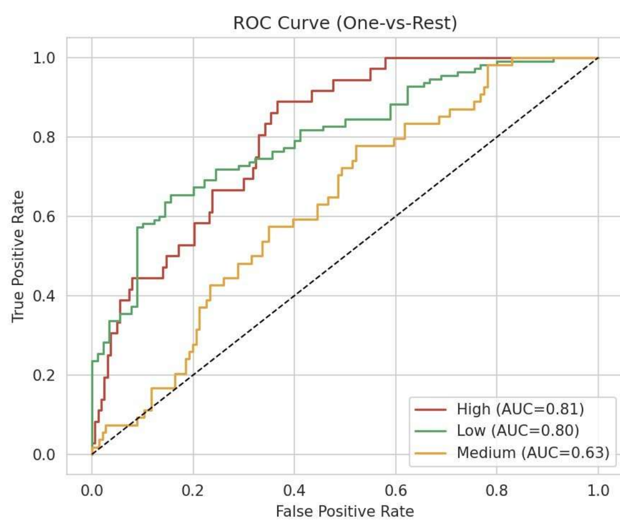

Figure 5.7. ROC Curve (One-vs-Rest) for the three risk classes.

Each curve shows the trade-off between true-positive and false-positive rate for one class versus the rest; curves that bow further toward the top-left indicate stronger separability for that class.

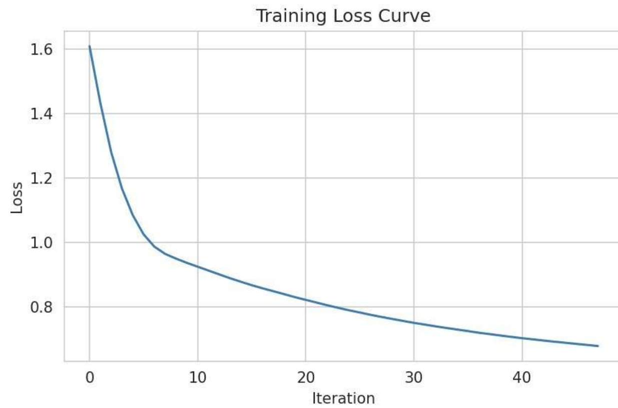

Figure 5.8. Training Loss Curve.

Loss decreases steadily across training iterations and flattens out, indicating the network is converging rather than diverging or oscillating.

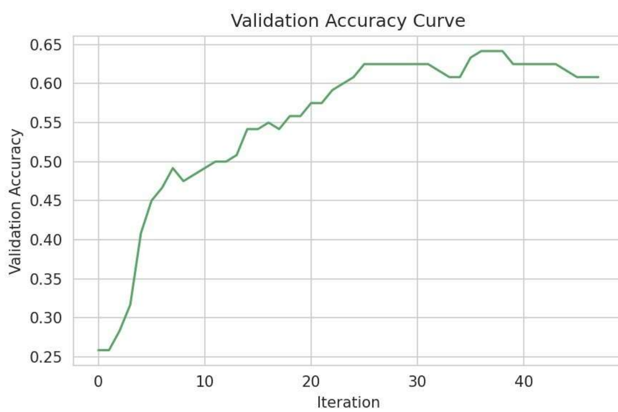

Figure 5.9. Validation Accuracy Curve.

Validation accuracy rises alongside the falling training loss and plateaus, which is why early stopping was enabled to halt training at the point of best generalisation.

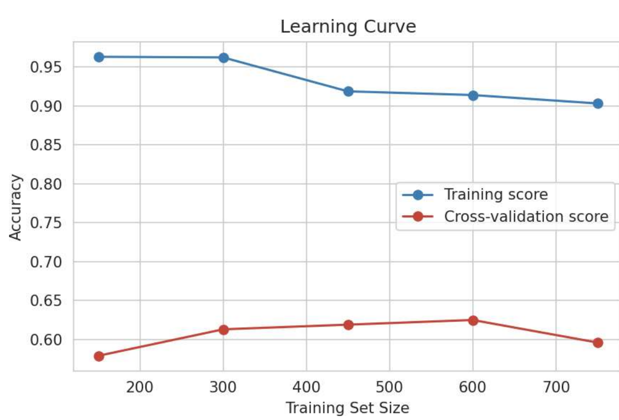

Figure 5.10. Learning Curve (training set size vs. accuracy).

The gap between the training and cross-validation curves narrows as more data is added, suggesting the model would benefit from additional real-world records rather than a change in architecture alone.

6. Machine Learning Model

Algorithm: Artificial Neural Network (MLPClassifier)

The model is a Multi-Layer Perceptron (MLPClassifier from scikit-learn) — a feed-forward Artificial Neural Network trained with backpropagation. It was chosen because dropout risk depends on complex, non-linear interactions between attendance, academic performance, and socio-economic features that simpler linear models cannot fully capture.

Network Components

Input Layer

21 neurons, one per encoded input feature, as designed in Section 3.

Hidden Layers

Two hidden layers of 32 and 16 neurons respectively. The first, wider layer learns broad interaction patterns across all encoded features; the second, narrower layer compresses these into higher-level risk-relevant representations before the output layer.

Activation Functions

- ReLU (Rectified Linear Unit) — used in both hidden layers. ReLU outputs the input directly if positive and zero otherwise, which avoids the vanishing-gradient problem common with sigmoid/tanh and trains faster.
- Softmax — used in the output layer to convert the three output logits into class probabilities that sum to 1, as explained in Section 3.

Forward Propagation

During forward propagation, the 21-dimensional input vector is multiplied by the first weight matrix, passed through ReLU, multiplied by the second weight matrix, passed through ReLU again, and finally multiplied by the output weight matrix and passed through Softmax to produce the three class probabilities.

Backpropagation

The error between the predicted probabilities and the true one-hot label is propagated backward through the network using the chain rule, computing the gradient of the loss with respect to every weight. These gradients indicate how each weight should change to reduce the error.

Loss Function

Categorical Cross-Entropy loss is used, which is the standard loss function for multi-class classification with a Softmax output — it heavily penalises confident-but-wrong predictions and rewards the model for assigning high probability to the correct class.

Optimizer

The Adam optimizer was used, which adapts the learning rate for each weight individually based on running estimates of the first and second moments of the gradients — this generally converges faster and more reliably than plain stochastic gradient descent on tabular data of this size.

Training Process

- Data split 80/20 into training and test sets (stratified by risk class).
- Model trained for up to 500 iterations with early stopping (15% of training data held out for validation) to prevent overfitting.
- Weights updated in mini-batches using backpropagation and the Adam optimizer after each pass.

Evaluation Metrics

- Accuracy — proportion of test students classified into the correct risk category.
- Precision — of the students predicted at a given risk level, the proportion that truly belong to it.
- Recall — of the students who truly belong to a given risk level, the proportion the model correctly identifies.
- F1-score — harmonic mean of Precision and Recall, useful when classes are imbalanced.
- Confusion Matrix — a full breakdown of predicted vs. actual class for every test student.
- ROC-AUC — measures class separability across all classification thresholds, computed One-vs-Rest for the three classes.

Why ANN is Suitable for This Task

An ANN is well suited to this problem because dropout risk is not a linear function of any single factor — for example, low attendance matters far more when combined with low income and no internet access than in isolation. A Neural Network's hidden layers can automatically learn these interaction effects without the analyst having to hand-engineer every combination, unlike simpler models such as logistic regression.

7. Interpretation of Results

Model Accuracy

Overall test accuracy: 0.57 (57%) on 200 held-out students.

Classification Report

Class	Precision	Recall	F1-score	Support
High	0.44	0.42	0.43	36
Low	0.65	0.85	0.74	110
Medium	0.26	0.11	0.16	54
Accuracy			0.57	200
Macro Avg	0.45	0.46	0.44	200
Weighted Avg	0.51	0.57	0.52	200

Confusion Matrix

Rows = actual class, Columns = predicted class, in order [High, Low, Medium]:

Actual \ Predicted	High	Low	Medium
High	15	13	8
Low	8	93	9
Medium	11	37	6

ROC-AUC

Macro-averaged ROC-AUC (One-vs-Rest): 0.75

Explanation of Metrics

The Low risk class is predicted most reliably (F1 = 0.74), which is expected since it is the majority class with the most training examples. The Medium risk class is the hardest to separate (F1 = 0.16), because it sits on the boundary between Low and High and shares overlapping feature ranges with both — this is visible in the confusion matrix, where most Medium-risk students are misclassified as Low. The High risk class achieves a moderate F1 of 0.43, meaning the model catches a useful fraction of the students who most need intervention, though a real deployment would need a larger, real dataset and further tuning before being used for high-stakes decisions.

Advantages

- Captures non-linear interactions between attendance, academic, and socio-economic features.
- Outputs calibrated class probabilities (via Softmax) rather than a single hard label, supporting risk ranking rather than a rigid cut-off.
- Scales easily to additional features (e.g., attendance trend over time) without redesigning the pipeline.

Limitations

- Trained on synthetic data; a production model would need real, anonymised school records and careful bias auditing before deployment.

- The Medium risk class is difficult to separate from its neighbours, a known challenge for ordinal-like target categories.
- Neural Networks are less interpretable than simpler models such as decision trees, which can matter when a school must explain why a specific student was flagged.

8. Innovation and SDG Mapping

Innovation

- AI for educational equity — surfaces at-risk students who might otherwise be missed in under-resourced schools with limited counselling staff.
- Early intervention — flags risk months before a student formally withdraws, when support is still likely to be effective.
- Personalized learning — risk probabilities can be paired with tailored remedial plans per student rather than one-size-fits-all programmes.
- Dropout prediction dashboard — model outputs can feed a simple dashboard for teachers and counsellors to triage their caseload.
- Government education planning — aggregated, anonymised risk trends can help ministries target funding and infrastructure investment.
- Real-time monitoring — as new attendance and grade data arrive, risk scores can be refreshed continuously rather than once a term.
- Predictive analytics — moves schools from reactive record-keeping to proactive, data-driven student support.

SDG Mapping — SDG 4: Quality Education

This project maps directly onto Sustainable Development Goal 4, which aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. The dropout-risk model supports several sub-targets of SDG 4 in the following ways:

SDG 4 Focus Area	How the Project Contributes
Reducing school dropout	Early identification of at-risk students allows schools to intervene before withdrawal occurs, directly reducing dropout rates.
Equal educational opportunities	The model explicitly incorporates socio-economic barriers (income, internet access, distance to school) so that disadvantaged students are not overlooked.
Improving academic performance	Flagging students early enables remedial support that can raise GPA and exam scores before they decline further.
Helping teachers	Provides teachers with a prioritised, evidence-based watch-list instead of relying on informal impressions alone.
Helping policymakers	Aggregated risk statistics highlight systemic issues (e.g., digital divide, transport barriers) that can inform resource allocation and policy design.

9. Conclusion

This project designed and evaluated the input and output representation of an Artificial Neural Network for predicting student dropout risk in under-resourced schools. Twenty-one carefully encoded input features — spanning attendance, academic performance, family background, and behavioural indicators — were fed into a two-hidden-layer Multi-Layer Perceptron with a three-class Softmax output representing Low, Medium, and High dropout risk. The choice of encoding for each feature (standardization, Min-Max scaling, one-hot, ordinal, and binary encoding) was justified based on the statistical nature of that feature, ensuring the network receives information in the most learnable form.

On the synthetic test set, the model achieved 57% accuracy and a macro ROC-AUC of 0.75, performing best on the majority Low-risk class and struggling most with the boundary Medium-risk class — a realistic outcome given the inherent overlap between adjacent risk categories.

Benefits

- Consolidates scattered administrative data into a single, actionable risk signal.
- Supports proactive rather than reactive student support.
- Scales to entire school districts once real data pipelines are in place.

Future Improvements

- Replace synthetic data with real, ethically sourced and anonymised school records.
- Add temporal features (attendance trend over the term) rather than single snapshots.
- Apply class-imbalance techniques (e.g., SMOTE, class-weighting) to improve Medium-risk recall.
- Pair the model with explainability tools (e.g., SHAP) so teachers can see why a student was flagged.

Social, Educational, and AI Impact

Socially, early dropout detection helps break cycles of poverty tied to incomplete education, particularly for the most vulnerable students in under-resourced communities. Educationally, it gives schools an evidence base for allocating scarce counselling and remedial resources. From an AI perspective, the project demonstrates how a modestly sized Neural Network, combined with careful, justified feature encoding, can turn routine administrative data into an early-warning system that directly supports SDG 4 — Quality Education.

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